

Olist E-commerce Dataset Analysis

Tools Used :- Python, Mysql,Powerbi

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1. Introduction

The Olist dataset is a real-world Brazilian e-commerce dataset that has around 1 lakh orders from 2016 to 2018. I analyzed this data to understand how customers behave, how deliveries are performing, and what payment methods people prefer. For this analysis I used Python for cleaning and merging the data, MySQL for running business queries, and Power BI for making the dashboard.

Key Findings at a Glance :-

1. Late deliveries have a strong and statistically significant negative impact on review scores.
2. Credit card users place significantly higher value orders compared to boleto users.
3. Surprisingly, product price has no significant effect on customer review scores.
4. 93.5% of orders are delivered on time — but remote northern states face 3x longer delivery times.
5. Health & Beauty is the top revenue-generating category across the platform.

2. Exploratory Data Analysis (EDA)

2.1 Data Cleaning & Merging (Python)

All 9 CSV files were loaded into Python using Pandas. The tables were merged using left joins on order_id, customer_id, and product_id as common keys. Date columns were converted to datetime format, and two new columns were made :

1. delivery_days: number of days from purchase to delivery
2. delay_days: difference between actual and estimated delivery date (positive = late)

After cleaning I kept only the delivered orders which gave me 115,715 rows and 41 columns to work with. For missing category names I filled them as 'unknown' and for missing review scores I used the median value.

2.2 Business Queries (MySQL)

After cleaning, the dataset was exported to MySQL for business-level analysis. The following 8 queries were executed:

Query	Key Finding
1. Top 10 Categories by Revenue	Health & Beauty leads with highest revenue
2. Payment Type Distribution	Credit card dominates at 76.8%
3. Review Score Distribution	Score 5 is most common rating
4. Top 10 States by Orders	SP accounts for majority of orders
5. Avg Delivery Days by State	RR, AP, AM have worst delivery (25+ days)
6. Repeat Purchase Rate	97% of customers never returned
7. Avg Order Value by Payment	Credit card users spend the most
8. Late vs On-Time Delivery	93.5% orders delivered on time

2.3 Dashboard Insights (Power BI)



Top Categories by Revenue: Health & Beauty leads all categories, followed by Watches & Gifts and Bed & Bath. These three categories contribute a significant portion of total platform revenue.

Geographic Distribution: SP has the highest revenue of around 6M, which is way more than other states. This shows that most business is coming from big cities and smaller states still have a lot of room to grow.

Delivery Performance by State: Remote northern states (RR, AP, AM) average 25-28 delivery days vs SP's 8 days.

Payment Behavior: 76.8% of customers use credit cards. Boleto accounts for 19.9%, while vouchers and debit cards are minimal.



Repeat Purchase Rate: One surprising finding was that 97% of customers only bought once and never came back. This is a big problem for the business because getting new customers costs money but keeping old ones is cheaper.

3. Hypotheses & Statistical Testing

Hypothesis 1: Late deliveries lead to lower review scores

Test Used :- T-test

T-statistic :- 125.78

P-value :- 0.0

Result :- VALIDATED

The T-test result is very strong with P-value of 0.0 which means this result is not by chance at all. Customers who got their orders late gave much lower ratings compared to customers who got orders on time. This was the most important finding in my whole analysis.

Business Insight: Late deliveries are directly hurting customer ratings. The company should focus on improving delivery speed especially in states like RR and AP where average delivery time is more than 25 days.

Hypothesis 2: Credit card users place higher value orders than boleto users

Test Used :- T-test

T-statistic :- 15.26

P-value :- $8.01e-53$ (near zero)

Result :- VALIDATED

Credit card users spend significantly more per order than boleto users. The near-zero P-value confirms this difference is statistically significant and not due to chance.

Business Insight: Since credit card users spend more, marketing team should target them for expensive products because they are more willing to pay.



Hypothesis 3: More expensive products receive better reviews

Test Used :- Independent T-test

Median Price :- 74.90

Avg Review (Cheap) :- 4.08

Avg Review (Expensive) :- 4.09

T-statistic :- 1.58

P-value :- 0.057

Result :- NOT VALIDATED

This result surprised me. Both cheap and expensive products got almost the same average rating - 4.08 vs 4.09. Since P-value is 0.057 which is above 0.05 threshold, this hypothesis was not validated.

Business Insight: Price alone does not make customers happy. What matters more is whether the product was delivered on time and matched what was shown. So the company should focus more on delivery experience and product quality instead of just pricing.

4. Actionables

Actionable 1: Send Alert to Customers for Late Orders

From the data I found that states like RR, AP and AM have very high delivery time of more than 25 days. When orders get late, customers give bad ratings. So the company should send a message to customers when their order is likely to get delayed. A small discount on next order can also be given to keep the customer happy and avoid bad reviews.

Actionable 2: Bring Back One Time Buyers

One of the most surprising findings was that 97% of customers only bought once and never came back. This means the company is losing a lot of repeat business. To fix this, the company should send a follow up email or notification to customers within 30 days of their first purchase. The message should show them products related to what they already bought. Credit card users should be targeted first because they spend more money per order.

5. Further exploration

Why do some product categories get bad reviews even when delivered on time?

From my analysis I found that late delivery is the main reason for bad reviews. But I think there might be some product categories that still get bad reviews even when the order was delivered on time. I want to explore this further to find out if the reason is bad product quality, wrong product photos or descriptions, or poor seller behavior. Finding this out would help the platform identify bad sellers and improve the overall customer experience.



6. LLM Usage Discloser

the following discloses exactly where AI assistance was used in this project:

1. For python syntax claude ai was used for guidance on pandas merge syntax and datetime conversion.
2. For Data cleaning also claude ai was used for suggestions for handling null values and filtering delivered orders.
3. For statistical test also claude ai was used for guidance on when to use T-test vs correlation and getting the syntax right.
4. For code debugging also claude ai was used .
5. Slight help in report making for getting report structure claude was used rest of report written by me.

All analysis decisions, hypotheses, business insights, observations, and report writing were done independently. LLMs were used for technical syntax help and debugging.